

Antipsychotics

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Session Objectives

- 1. Distinguish between the positive and negative symptoms of schizophrenia.**
- 2. Describe the typical antipsychotics, with respect to their receptor actions. How does this contribute to the adverse effects of these drugs?**
- 3. Describe these adverse effects and ways to manage the patient in a clinical setting.**
- 4. Describe the atypical antipsychotics. How do they differ in mechanism and adverse effects to typical antipsychotics?**
- 5. Describe other clinical conditions that can be treated with antipsychotics.**

Definition

Schizophrenia and psychosis are categorized:

- Positive symptoms: delusions, agitation, hallucinations, disorganized thinking “**typicals**” **Mesolimbic**
- Negative symptoms: apathy, social isolation, poor speech “**atypicals**” **Mesocortical**

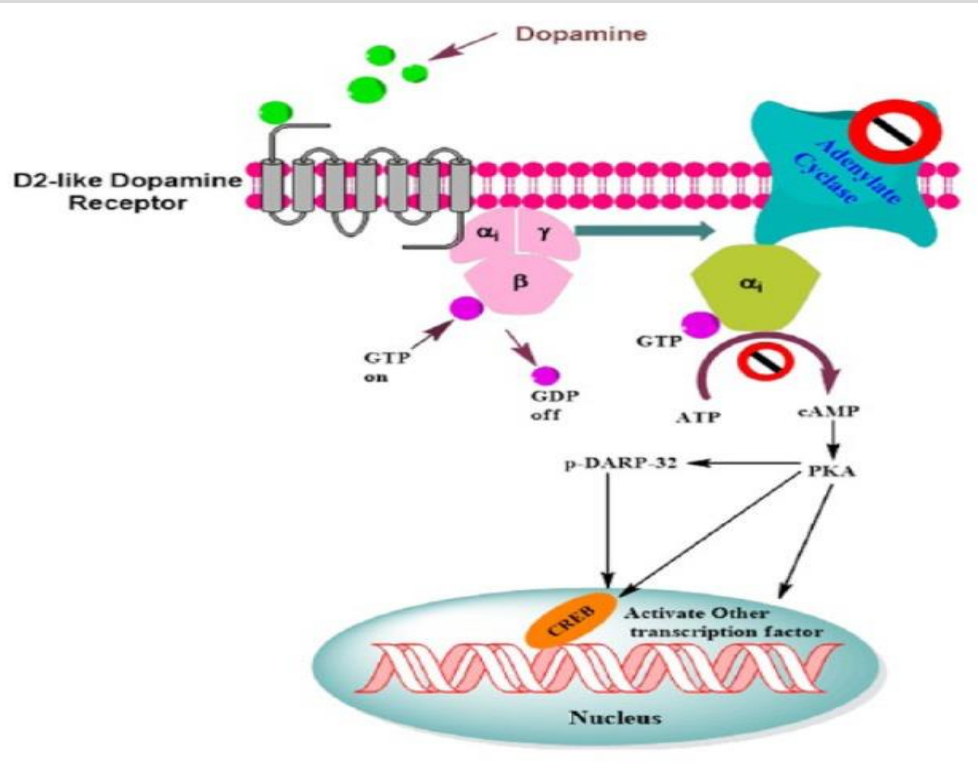
Dopamine Hypothesis

- **Excess of dopamine in mesolimbic system**
- **Increased D₂ receptors postmortem in untreated schizophrenics**
- **Increased urinary homovanillic acid (HVA)**
- **Drugs that increase dopamine (amantadine) or inhibit reuptake (cocaine) induce psychotic behavior.**

D₂ receptors

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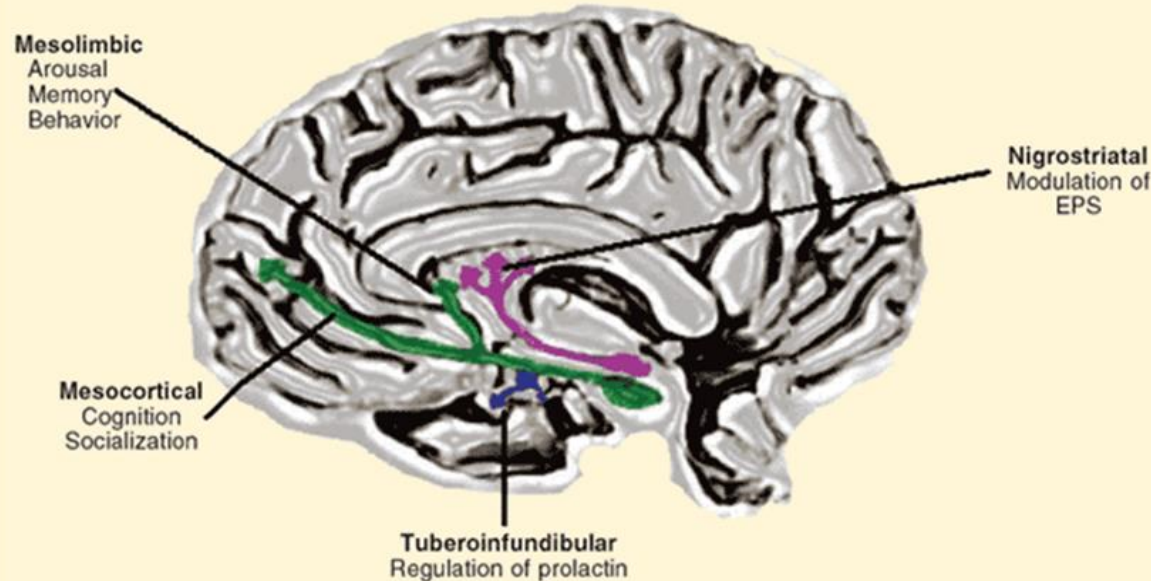


Dopamine Pathway

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Figure 2. Dopamine System Pathways and Clinical Function



Paquet D (graphic design). The Brain from Top to Bottom. Canadian Institutes of Health Research: Institute of Neuroscience, Mental Health and Addiction. http://thebrain.mcgill.ca/flash/lala_03/la_03_cl/la_03_cl_que/la_03_cl_que.html. Accessed February 23, 2008. Figure is under copyleft and has been adapted by Paul D. Mergenhagen.

Source: Course Syllabus

Typical Antipsychotics (First Generation)

- Blockade of $D_2 \ggg 5-HT_2$ (mesolimbic)
- Blockade of both presynaptic and postsynaptic D_2 receptors

But:

- Blockade of D_2 in mesocortical pathway (worsens negative symptoms)
- Blockade of D_2 in nigrostriatal pathway (EPS, Parkinsonism)
- Blockade of D_2 in tuberoinfundibular pathway (prolactin release)

Typical Antipsychotic Indications

- **Schizophrenia**
- **Drug-induced psychosis**
- **Manic phase of bipolar disorder (mainly atypicals)**
- **Dementia (also atypicals) ?**
- **Antiemetics, gastroparesis (metoclopramide)**
- **Tourette's (haloperidol, pimozide)**

Drugs

- **Chlorpromazine**
- **Fluphenazine**
- **Haloperidol**
- **Thioridazine**
- **Loxapine**
- **Trifluoperazine**
- **Thiothixene**
- **Pimozide**

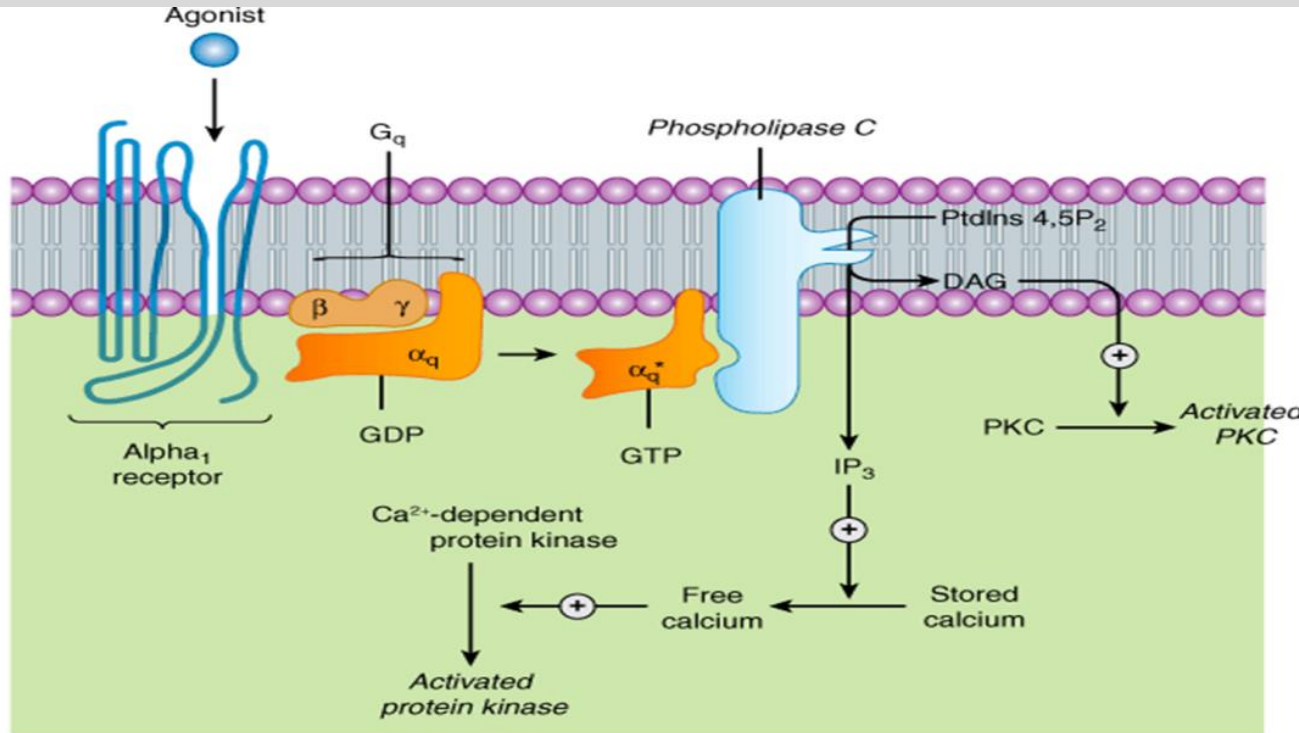
Typical Antipsychotic Adverse effects

- M, H₁, alpha₁ blockade
- EPS (dystonia, pseudoparkinsonism)
- Akathisia
- Menstrual irregularities (prolactin release)
- Tardive dyskinesia
- Neuroleptic malignant syndrome (NMS)

Gq: 5-HT_{2A}, Alpha₁, M₃, H₁ V₁

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Source: Course Syllabus

Tardive dyskinesia

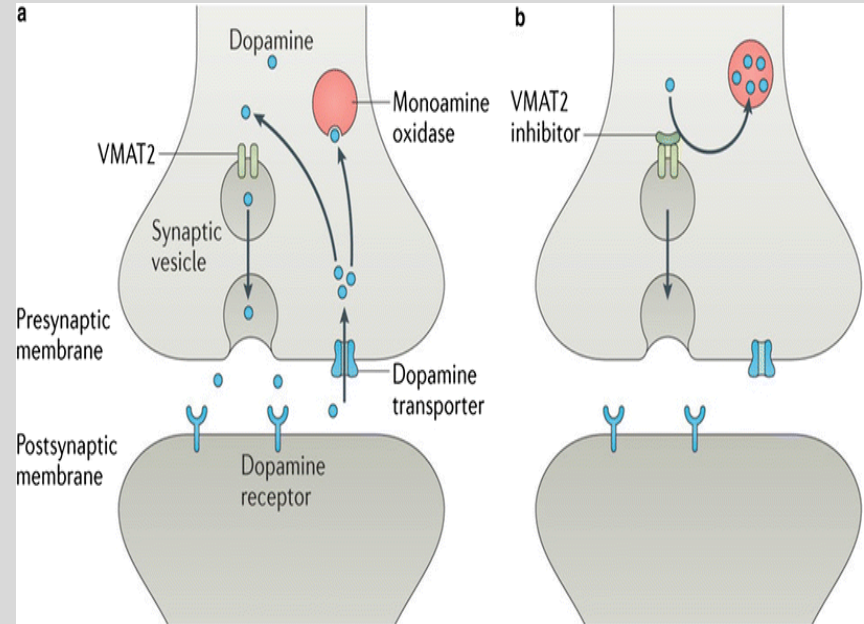
- D₂ blockade will up-regulate receptors
- Occurs months to years later
- Tongue protrusion, lip-smacking, abnormal movements. “rabbit syndrome”
- Start with lowest dose of drug, assess patient periodically. Use atypicals.

Management of Tardive dyskinesia

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- **Vesicular monoamine transporter-2 (VMAT₂) inhibitors: valbenazine, deutetrabenazine.**
- **Also used for Huntington disease.**
- **Related to reserpine**
- **Adverse effects: Increased QT, sedation, substrate for CYP**



Neuroleptic Malignant Syndrome

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- Muscle rigidity, loss of consciousness, tachycardia, high fever, diaphoresis, elevated creatine kinase
- Occurs weeks to months later (with no GI effects), unlike serotonin syndrome.
- Discontinue drug, provide supportive care, and provide dantrolene and bromocriptine to treat NMS
- Use atypical drugs in future

NMS vs Serotonin Syndrome

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	NMS	Serotonin Syndrome
Onset	Gradual (months-weeks)	Abrupt
Findings	“Lead-pipe” rigidity	Myoclonus and tremor
Reflexes	Decreased	Increased
Pupils	Normal	Mydriasis
Drugs	Antipsychotics, halogenated anesthetics	Linezolid, St. John’s Wort, “triptans”, SSRIs, MAOIs, tricyclic antidepressants

Source: Course Syllabus

Dantrolene

- **Muscle relaxant**
- **Binds to the ryanodine receptor, prevents calcium release from the sarcoplasm.**
- **Used IV for NMS and the toxicity associated with halogenated anesthetics (malignant hyperthermia).**

Specific Adverse Effects

- Chlorpromazine, Thioridazine (low potency, most muscarinic, histamine, and α_1 blockade). Highest QT interval increase with thioridazine
- Fluphenazine, haloperidol (high potency, most D_2 blockade)
- Thioridazine (pigment retinopathy)
- Chlorpromazine (corneal deposits)

Management of Adverse Effects

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- Anticholinergic (**benztropine, diphenhydramine**) for dystonia and parkinsonism (**Amantadine**)
- Akathisia (**propranolol**)
- NMS (**dantrolene**)

Muscarinic Antagonists

- **Benztropine, trihexyphenidyl**
- **Helps with tremor associated with D₂ antagonists**
- **Name the adverse effects ??**

Serotonin Hypothesis

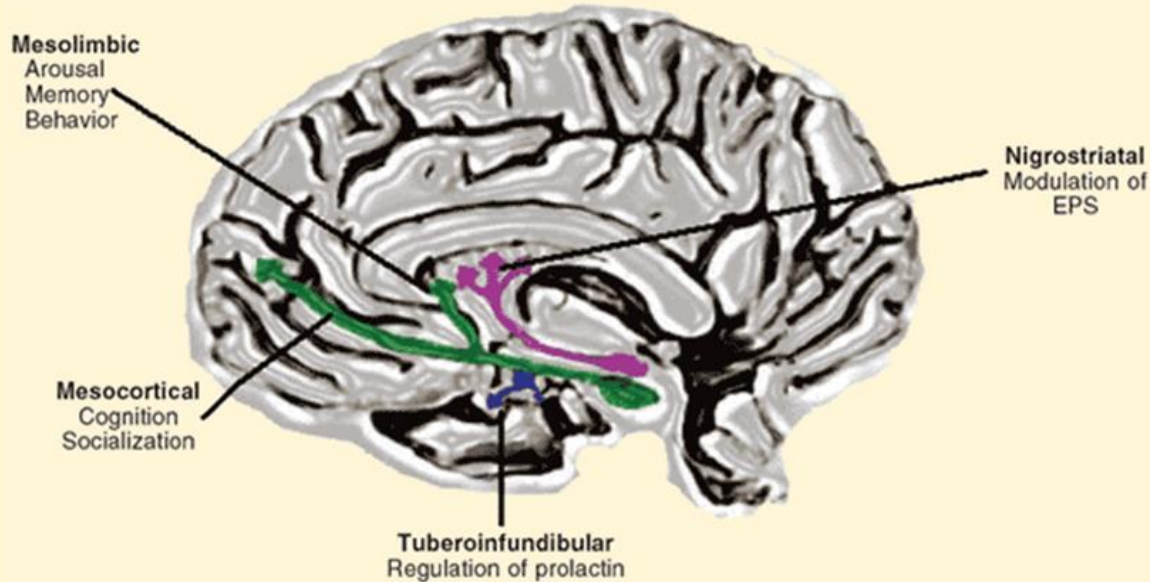
- **LSD mimics schizophrenia by stimulating 5-HT_{2A} receptor**
- **5-HT_{2A} block in tuberoinfundibular pathway decreases release of prolactin.**
- **Mesocortical pathway: Block of 5-HT_{2A} increases dopamine (improves negative symptoms).**
- **Mesolimbic pathway: Block of 5-HT_{2A} does not increase dopamine levels**

Dopamine Hypothesis

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Figure 2. Dopamine System Pathways and Clinical Function



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Atypical Antipsychotics (Second generation)

- 5-HT₂ >>>D₂; D₃ and D₄ block
- Increase DA release to relieve negative symptoms of psychosis
- Fewer extrapyramidal effects (EPS) and hyperprolactinemia
- **All prolong QT interval. Change in metabolic panel.**
- Aripiprazole, olanzapine, quetiapine, risperidone, ziprasidone, and cariprazine, approved for mania

Clozapine

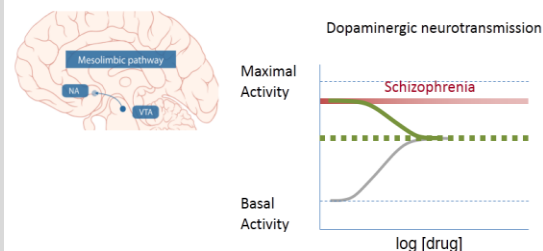
- First atypical drug. For treatment resistant schizophrenia.
- Blockade of $D_4 = \alpha_1 > 5\text{-HT}_{2A} > D_2 = D_1, H_1$ also
- Night-time “drooling” (paradox) M_1 and M_4 receptor agonism?
- Weight gain (block 5-HT_{2c})? Locaserin (agonist of 5-HT_{2c} , former use for weight loss). Also, H_1 block.
- Hypotension
- Agranulocytosis: do weekly blood counts for the first 6 months of treatment. Every 2 weeks from 6-12 months, then monthly.

Aripiprazole

- Partial D_2 and $5-HT_{1a}$ agonist, $5-HT_{2A}$ antagonist.
- Used in bipolar disease, schizophrenia, Tourette's and irritability in autism.
- $5-HT_{2C}$ agonist (minimal weight gain).
- CYP substrate (3A4, 2D6)
- Brexpiprazole: Less D_2 intrinsic agonist activity than aripiprazole. More potent action at $5-HT_{1a}$ and $5-HT_{2A}$.
 - Manages dementia agitation

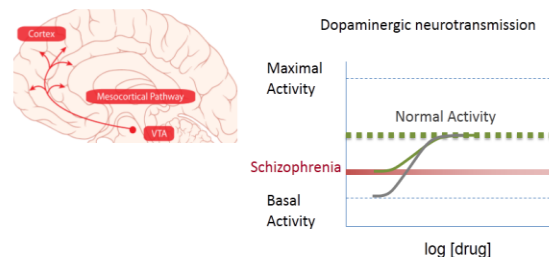
Source: Course Syllabus

Aripiprazole therapy



Courtesy of Richard Mailman, modified from Mailman RB, Murthy V. Current pharmaceutical design 2010;16:488-501

Aripiprazole Therapy



Courtesy of Richard Mailman, modified from Mailman RB, Murthy V. Current pharmaceutical design 2010;16:488-501.

Atypical Antipsychotics

- Olanzapine: Less ANS effects, Blocks D_3 , D_4 , Fewer EPS, **weight gain**, sedation. Approved in combination with fluoxetine for mania.
 - Block $5-HT_{2A} > H_1 > D_4 > D_2 > \alpha_1 > D_1$
- Quetiapine: D_2 blocker, but binds for a short time. **Minimal muscarinic block**, but H_1 and α_1 . **Most sedation.**
 - Block $H_1 > \alpha_1 > M_{1,3} > D_2 > 5-HT_{2A}$

Atypical Antipsychotics

- **Risperidone:** Increase prolactin, hyperlipidemia, hyperglycemia
 - H₁ blockade (sedation, weight gain)
 - Alpha₁ block (hypotension)
- **Paliperidone:** active metabolite of risperidone (24-hour oral dosing)
- **Ziprasidone:** blocks 5-HT and NE reuptake.
 - Skin reaction, eosinophilia
 - H₁ blockade (sedation, weight gain)
 - Alpha₁ block (hypotension)
- **Cariprazine:** D₂, D₃ partial agonist (5-HT_{1A} agonist)

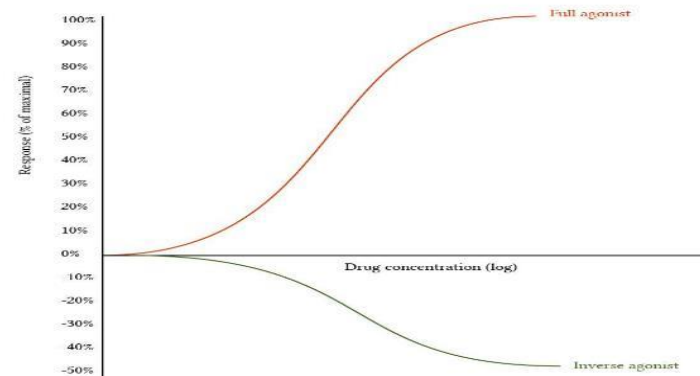
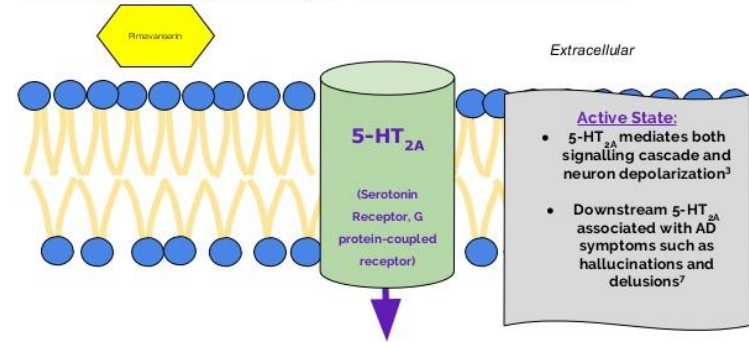
Pimavanserin (Nuplazid®)

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- Inverse agonist of 5-HT_{2A} receptors
- No D₂, 5-HT_{2B}, M, H₁ or alpha₁ action.
- Use: Parkinson's, Alzheimer's patient with psychosis.

Source: Course Syllabus

Alzheimer's Disease (AD) - Mechanism of Action

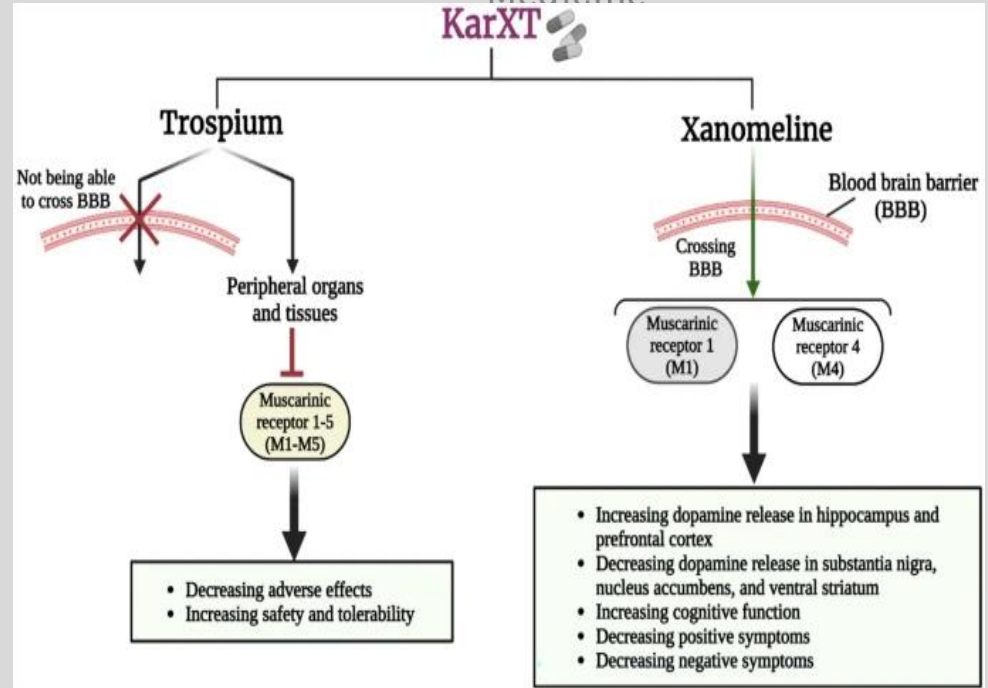


Xanomeline and trospium chloride (Cobenfy[®])

- First antipsychotic targeting cholinergic receptors
- Xanomeline: M₄, M₁ agonist
 - M₄ and M₁ receptors in limbic and cortex control cognition and affect
 - Overall: rebalance dopamine and glutamate
- Trospium: nonselective muscarinic antagonist; manages AE of muscarinic agonist. No cross through BBB
- Possible for AD psychosis??

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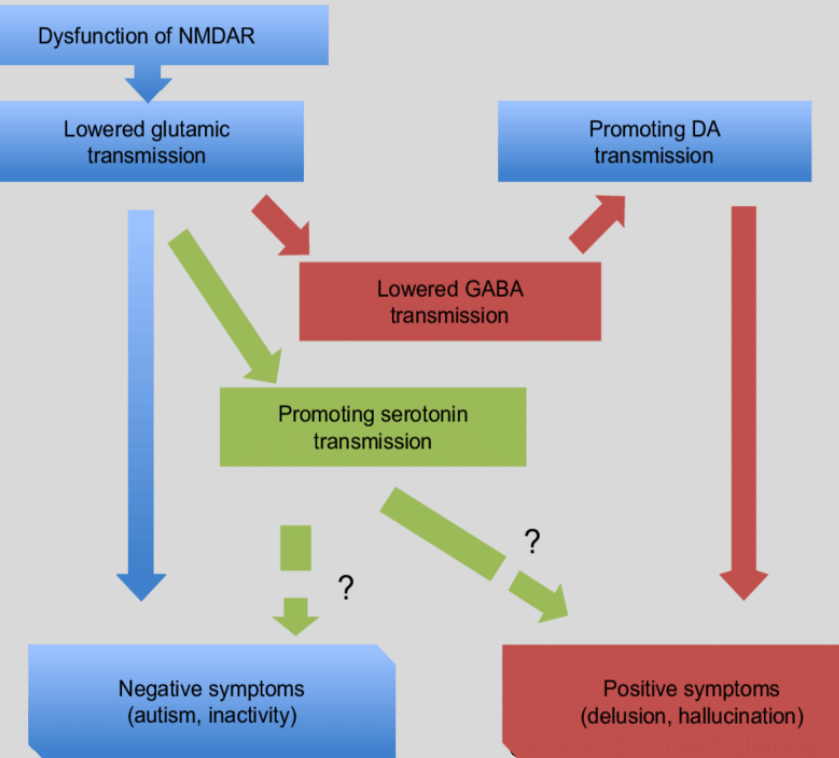


Lumateperone (Caplyta®)

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Glutamate hypothesis of schizophrenia



- 5-HT_{2A} receptor antagonist (60x greater affinity for this receptor vs D₂ receptors).
- D₂ receptor presynaptic partial agonist and postsynaptic antagonist “Balance”
- D₁ receptor–dependent modulator of glutamate.

Summary

Adverse effects profile

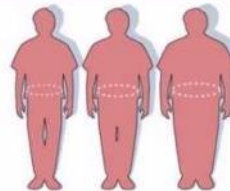
First Generation Antipsychotics

- Higher risk of neurological side effects



Second Generation Antipsychotics

- Higher risk of metabolic side effects



Summary

- **Describe the typical antipsychotics, with respect to the receptor actions. How does this contribute to the adverse effects of these drugs?**
- **Describe the atypical antipsychotics. How do they differ in mechanism and adverse effects to typical antipsychotics?**

Core References:

1. **Basic & Clinical Pharmacology, 16e. Bertram G. Katzung, Todd Vanderah, Chapter 29, Antipsychotic agents**
2. **Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e. Chapter 19, Pharmacotherapy of Psychosis and Mania.**
3. **Antipsychotics summary guide and practice questions (Maria A. Pino, Ph.D).**

Lecture Feedback Form:

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>